

KUMARAGURU
Institutions

2025

**Micro
cosm**
The Co-existing Architecture

Carbon Emissions Report 2025

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Introduction

Kumaraguru Institutions is a multi-campus educational institution in Tamil Nadu, South India, offering undergraduate, postgraduate, and research programs in Engineering, Technology, Management, Agriculture, Liberal Arts, and Science. Guided by its motto, "Character is Life," the institution is committed to academic excellence, holistic development, and nurturing responsible individuals with integrity and purpose. Through experiential learning, innovative research, and strong community engagement, Kumaraguru fosters a culture of inquiry and social responsibility, empowering students to address real-world challenges and contribute meaningfully to society.



Microcosm is a long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance, resource efficiency, and responsible environmental stewardship.

The scope of this report covers all major operational aspects, including energy consumption, transportation, and direct emissions from stationary sources. By quantifying these impacts, we lay the groundwork for targeted reduction strategies and continuous improvement.

Introduction

Kumaraguru Institutions recognize the imperative to address climate change and integrate environmental sustainability into campus operations. This report encapsulates our Greenhouse Gas (GHG) inventory, providing a transparent account of our carbon footprint and resource utilization.

The scope of this report covers all major operational aspects, including energy consumption, transportation, and direct emissions from stationary sources. By quantifying these impacts, we lay the groundwork for targeted reduction strategies and continuous improvement.

Strategic Objectives

- Establish a robust baseline for GHG emissions and energy consumption.
- Identify key emission hotspots to prioritize mitigation interventions.
- Monitor the effectiveness of ongoing sustainability initiatives, such as renewable energy integration.

- Foster a culture of environmental stewardship among students, faculty, and staff.

Highlights



EMISSION PROFILE

Total GHG Emissions

1753.80 tCO₂e

Emissions (DG + Transport) **389.1** tCO₂e

Emissions (Electricity) **1364.7** tCO₂e

Emissions (LPG - Canteen) **0.00** tCO₂e



ENERGY INDICATORS

Total Energy Consumption

5250515 kWh

Grid Energy Consumption **1877147** kWh

Renewable Energy Consumption **3373368** kWh

Renewable Energy Share **64.2** %

Transport Indicators

Fleet Fuel Consumption Overview



TRANSPORT INDICATORS (FLEET)

Total Fuel Consumption

126034 Litres

Total Petrol Consumption **8954.11** Litres

Total Diesel Consumption **117080** Litres

Total Fleet Emissions **337.6** tCO₂e

Climate Action

Sustainability Mitigation Efforts



CLIMATE ACTION INDICATORS

Avoided Emissions

2452.4 tCO₂e

Per Capita Emissions

0.000 tCO₂e/person

Executive Summary

High-level overview of carbon performance for Carbon Emissions Report - 2025.

Performance Overview

During this reporting period, the campus generated 1753.80 tCO₂e of gross carbon emissions. Through our renewable energy initiatives, we successfully avoided 2452.44 tCO₂e, resulting in a net carbon impact of -698.64 tCO₂e. We remain committed to our long-term decarbonization pathway.



CARBON STORY

Gross GHG Emissions

1753.8 tCO₂e

Net GHG Emissions

-698.6 tCO₂e

Avoided Emissions

2452.4 tCO₂e



STRATEGIC RECOMMENDATIONS

- Accelerate the transition of internal campus mobility to Electric Vehicles (EVs) to reduce Scope 1 transport emissions.
- Expand rooftop solar capacity to push the renewable energy mix beyond the current baseline.

- Implement automated BMS (Building Management Systems) for high-load commercial blocks to optimize HVAC consumption.

Scope 1 Story

Direct Emissions Overview

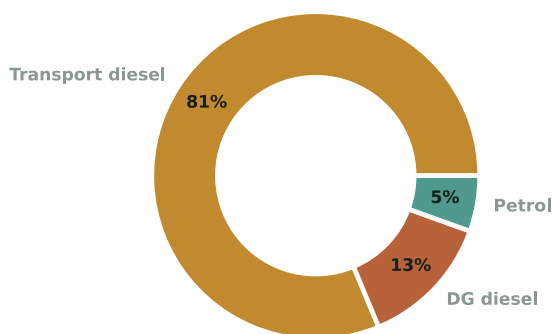
Scope 1 emissions generated 389.11 tCO₂e. Transport remains the dominant emission source in this category. Overall Scope 1 emissions decreased by 0% compared to the previous period.

Transport Deep Dive

Transport emissions decreased by 0%. Continued fleet optimization is contributing to this reduction.

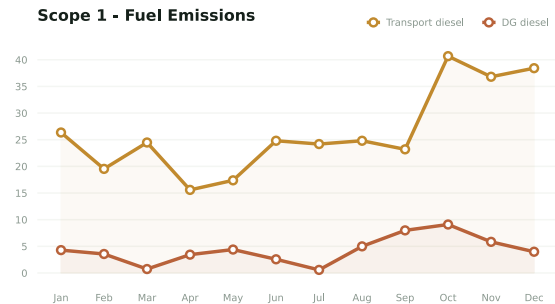
Scope 1 Composition

Scope 1 Breakdown



Emissions Trend

Scope 1 - Fuel Emissions



Transport Emissions (Scope 1)

337.6 tCO₂e

MONTH	PETROL (LITRES)	EMISSIONS (TCO ₂ E)	DIESEL (LITRES)	EMISSIONS (TCO ₂ E)
JAN	746.0	1.7814	9757	26.3537
FEB	483.5	1.1546	7230	19.5282
MAR	493.0	1.1773	9063	24.4792
APR	252.0	0.6018	5771	15.5875
MAY	440.0	1.0507	6435	17.3809
JUN	414.0	0.9886	9184	24.8060
JUL	853.0	2.0370	8955	24.1875
AUG	908.0	2.1683	9184	24.8060
SEP	1123.0	2.6817	8588	23.1962
OCT	976.2	2.3312	15060	40.6771
NOV	1124.6	2.6855	13629	36.8119
DEC	1140.8	2.7242	14224	38.4190

DG Set Emissions (Scope 1)

51.5 tCO₂e

MONTH	DIESEL (LITRES)	EMISSIONS (TCO ₂ E)
JAN	1588.62	4.2909
FEB	1319.34	3.5635
MAR	274.23	0.7407
APR	1273.80	3.4405
MAY	1628.22	4.3978
JUN	951.39	2.5697
JUL	214.17	0.5785
AUG	1849.32	4.9950
SEP	2959.11	7.9926
OCT	3368.64	9.0987
NOV	2162.82	5.8418
DEC	1475.43	3.9851

Scope 2 Story

Indirect Emissions Overview

Grid electricity consumption accounted for 1364.69 tCO₂e. This represents 77.8% of total gross emissions. Scope 2 emissions fell by 0%. Continuous efforts in optimizing HVAC and lighting can further reduce this footprint.

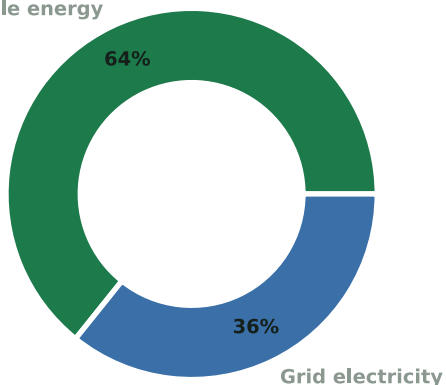
Renewable Impact

Renewable energy integration contributed 3373368.00 kWh, offsetting approximately 2452.44 tCO₂e of potential emissions. Our current renewable share stands at 64.2%.

Renewable Energy Mix

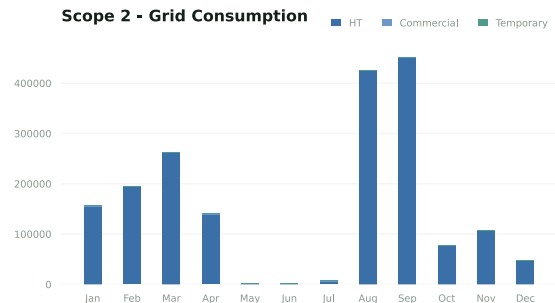
Renewable Mix

Renewable energy



Grid Consumption Profile

Scope 2 - Grid Consumption



Electricity Emissions (Scope 2)

1364.7 tCO₂e

RENEWABLE ENERGY

3373368 kWh

GRID ENERGY

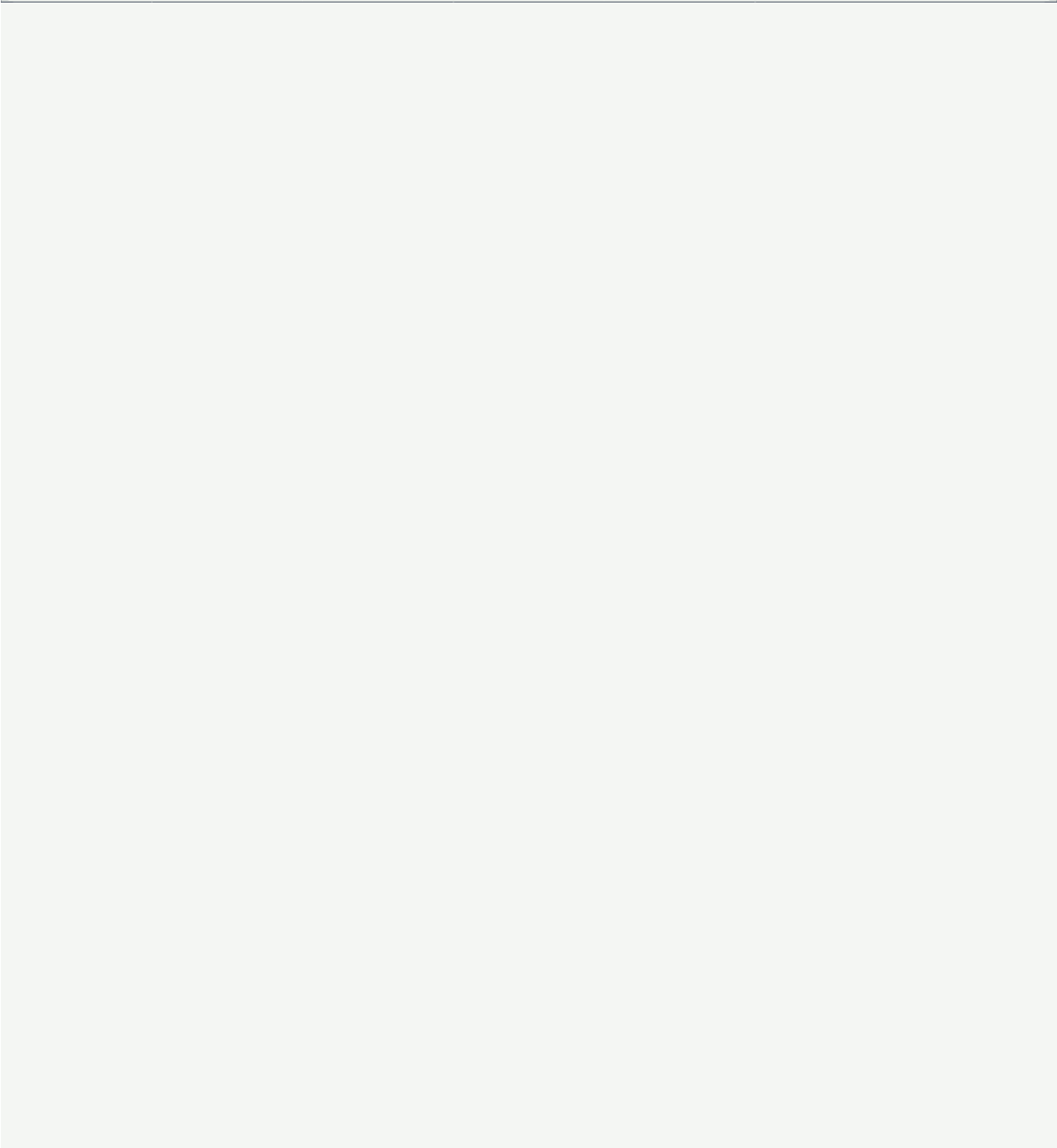
1877147 kWh

TOTAL CONSUMPTION

5250515 kWh

MONTH	GRID HT		GRID COMMERCIAL		GRID TEMPORARY	
	ELECTRICITY (KWH)	EMISSIONS (TCO ₂ E)	ELECTRICITY (KWH)	EMISSIONS (TCO ₂ E)	ELECTRICITY (KWH)	EMISSIONS (TCO ₂ E)
JAN	155222	112.846	1023	0.744	0	0.000
FEB	194516	141.413	862	0.627	0	0.000
MAR	261397	190.036	1119	0.814	0	0.000
APR	139071	101.105	1140	0.829	0	0.000
MAY	1321	0.960	1001	0.728	0	0.000
JUN	690	0.502	1023	0.744	0	0.000
JUL	5934	4.314	1128	0.820	428	0.311
AUG	424410	308.546	1094	0.795	416	0.302
SEP	450216	327.307	1027	0.747	389	0.283

MONTH	GRID HT		GRID COMMERCIAL		GRID TEMPORARY	
	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)	ELECTRICITY (KWH)	EMISSIONS (TCO2E)
OCT	76958	55.948	1000	0.727	346	0.252
NOV	105958	77.031	963	0.700	331	0.241
DEC	46972	34.149	899	0.654	293	0.213



Emissions Comparison

Performance analysis vs previous reporting period



CONSOLIDATED GHG SUMMARY

CATEGORY	CURRENT PERIOD	PREVIOUS PERIOD	% CHANGE
Total Emissions	1753.80	N/A	0.00%
Scope 1 Emissions	389.11	N/A	0.00%
Scope 2 Emissions	1364.69	N/A	0.00%
Renewable Energy (kWh)	3373368	N/A	0.00%

Progress Summary

The comparison table above illustrates our performance trajectory against the previous baseline. Reductions in total emissions are primarily driven by continuous optimization in energy usage and sustained renewable integration.

Methodology

Purpose of the Report

This report presents the Greenhouse Gas (GHG) emissions inventory for Kumaraguru Institutions for the reporting period, compiled in accordance with ISO 14064-1:2018. The objective of this inventory is to measure and disclose the institution's GHG emissions, and provide reliable data to support future emissions reduction strategies.

GHG Quantification Approach

The inventory has been developed using an activity-based quantification approach. Emissions were calculated by multiplying activity data (e.g., fuel consumption, electricity consumption) with appropriate emission factors derived from recognized national and international sources (IPCC, CEA, DEFRA).

Organizational Boundary

The organizational boundary has been defined using the operational control approach. All facilities and operations over which Kumaraguru Institutions has operational control are included. Scope 3 emissions have been assessed at a screening level but are excluded from quantitative reporting due to the unavailability of consolidated data.

Global Warming Potential (GWP)

All greenhouse gases were converted into carbon dioxide equivalent (CO₂e) using Global Warming Potential (GWP) values from the IPCC Sixth Assessment framework.

Conclusion

This Greenhouse Gas (GHG) inventory enhances the understanding of emission sources within campus operations and provides a foundation for tracking environmental performance and identifying opportunities for energy efficiency and emission reduction. The institution intends to progressively improve data quality, expand emission coverage, and integrate climate considerations into operational decision-making.

REPORT PREPARED BY

Microcosm Sustainability Intelligence

A long-term, campus-wide environmental sustainability initiative at Kumaraguru Institutions that aims to transform the institution into a model of ecological balance.